Xinyu Gao

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Summary

Ph.D. candidate in Mathematics at the University of Missouri working at the intersection of theoretical machine learning, generative models, and high-dimensional geometry. Current research focuses on diffusion model memorization, noisy data geometry, and theoretical aspects of generative modeling. Co-first author of an ICML 2026 paper on memorization in diffusion models.

Education

University of Missouri 2022 – Present
Ph.D. in Mathematics *Columbia, MO*

Shandong University 2017 – 2021
B.S. in Mathematics *Jinan, China*

Selected Publications

Two Calm Ends and the Wild Middle: A Geometric Picture of Memorization in Diffusion Models 2026
International Conference on Machine Learning (ICML 2026) *Co-first author*

- Developed a geometric framework explaining when diffusion models memorize training data across different noise scales.
- Introduced shell coverage and posterior concentration metrics to characterize memorization risk in diffusion training.
- Validated theoretical predictions through PyTorch experiments on CIFAR-10, CelebA, and diffusion finetuning settings.
- Demonstrated that reducing training emphasis in intermediate-noise regions decreases memorization while preserving generation quality.

Lebesgue Bounds for Multilinear Spherical and Lacunary Maximal Averages 2026
Journal of Fourier Analysis and Applications (JFAA) ***Xinyu Gao***

- Established new Lebesgue-space bounds for multilinear spherical and lacunary maximal operators.
- Developed harmonic analytic techniques for multilinear averaging operators and maximal estimates.

A.E. Convergence vs Boundedness 2026
arXiv:2602.16654 ***Xinyu Gao****, Loukas Grafakos*

- Studied relationships between almost-everywhere convergence and operator boundedness in harmonic analysis.

Research Experience

Research Assistant — Generative Models and Diffusion Theory 2025 – Present
University of Missouri *Advisor: Zhengchao Wan*

- Developed theoretical frameworks for understanding memorization phenomena in diffusion models and generative learning systems.
- Studied geometric properties of noisy high-dimensional data distributions and diffusion trajectories across noise scales.
- Co-authored an ICML 2026 paper introducing shell coverage and posterior concentration metrics for diffusion memorization analysis.

Graduate Researcher — Harmonic Analysis 2022 – Present
University of Missouri *Advisor: Loukas Grafakos*

- Conducted research on multilinear maximal operators, bilinear Hilbert transforms, and Fourier-analytic problems.
- Established new convergence and boundedness results in harmonic analysis and multilinear operator theory.
- Developed analytical techniques connecting maximal estimates, convergence theory, and geometric averaging operators.

Selected Projects

Diffusion Memorization Geometry 2026
PyTorch, CIFAR-10

- Built PyTorch evaluation tools for analyzing memorization behavior across 50+ diffusion checkpoints on CIFAR-10.
- Implemented shell coverage and posterior concentration metrics for studying intermediate-noise memorization effects.
- Developed nearest-neighbor retrieval and noisy trajectory visualization pipelines for diffusion model analysis.
- Created reproducible experimental workflows for generative model evaluation, logging, and visualization.

Diffusion and Flow Matching Implementations 2026
PyTorch

- Implemented DDPM and flow-matching models from scratch on low-dimensional and image datasets.
- Built modular training, sampling, and inference pipelines for generative modeling experiments.
- Implemented forward diffusion, denoising, and trajectory simulation components for controlled memorization studies.
- Explored score behavior and generation trajectories in simplified high-dimensional diffusion settings.

Technical Skills

Programming: Python, PyTorch, NumPy, Matplotlib, Git, Linux, Jupyter Notebook, LaTeX

Machine Learning: Diffusion Models, Flow Matching, Generative Modeling, High-Dimensional Data Analysis

Mathematics: Harmonic Analysis, Fourier Analysis, Probability Theory, Functional Analysis, High-Dimensional Geometry

Honors and Awards

Nigel Kalton Fellowship
University of Missouri *2025*

Mitacs Globalink Research Internship Recipient 2021

Academic Scholarship 2019
Shandong University

Teaching Experience

Teaching Assistant and Instructor for courses including Calculus I, Multivariable Calculus, Measure Theory, Advanced Calculus, and Undergraduate Mathematics Writing at the University of Missouri (2022–Present).